Appendix A

Advanced Calculus: A Review

This Appendix is a brief review of elementary set theory, real numbers, limits, sequences and series, continuity, differentiability, Riemann integration, complex numbers, exponential and trigonometric functions, and metric spaces. For proofs and further details, see Rudin (1976) and Royden (1988).

A.1 Elementary set theory

This section reviews the following: sets, set operations, product sets (finite and infinite), equivalence relation, axiom of choice, countability, and uncountability.

Definition A.1.1: A *set* is a collection of objects.

It is typically defined as a collection of objects with a *common defining property.* For example, the collection of even integers can be written as $\mathbb{E} \equiv \{n : n \text{ is an even integer}\}$. In general, a set Ω with defining property p is written as

$$\Omega = \{\omega : \omega \ \text{ has property } \ p\}.$$

The individual elements are denoted by the small letters ω, a, x, s, t, etc., and the sets by capital letters Ω, A, X, S, T, etc.

Example A.1.1: The closed interval $[0, 1] \equiv \{x : x \text{ a real number}, \ 0 \leq x \leq 1\}$.

Example A.1.2: The set of positive rationals $\equiv \{x : x = \frac{m}{n}, m$ and n positive integers$\}$.

Example A.1.3: The set of polynomials in x of degree 10 $\equiv \{P(x) : P(x) = \sum_{j=0}^{10} a_j x^j, a_j$ real, $j = 0, \ldots, 10, a_{10} \neq 0\}$.

Example A.1.4: The set of all polynomials in $x \equiv \{P(x) : P(x) = \sum_{j=0}^{n} a_j x^j, n$ a nonnegative integer, a_j real, $j = 0, 1, 2, \ldots, n\}$.

A.1.1 Set operations

Definition A.1.2: Let A be a set. A set B is called a *subset of* A and written as $B \subset A$ if every element of B is also an element of A. Two sets A and B are the *same* and written as $A = B$ if each is a subset of the other. A subset $A \subset \Omega$ is called *empty* and denoted by $\emptyset$ if there exists no ω in Ω such that $\omega \in A$.

Using the mathematical notation $\in$ and $\Rightarrow$, one writes

$$B \subset A \quad \text{if} \quad x \in B \Rightarrow x \in A.$$

Here '$\in$' means "belongs to" and $\Rightarrow$ means "implies."

Example A.1.5: Let $\mathbb{N}$ be the set of natural numbers, i.e., $\mathbb{N} = \{1, 2, 3, \ldots\}$. Let $\mathbb{E}$ be the set of even natural numbers, i.e., $\mathbb{E} = \{n : n \in \mathbb{N}, n = 2k$ for some $k \in \mathbb{N}\}$. Then $\mathbb{E} \subset \mathbb{N}$.

Example A.1.6: Let $A = [0, 1]$ and B be the set of x in A such that $x^2 < \frac{1}{4}$. Then $B = \{x : 0 \leq x < \frac{1}{2}\} \subset A$.

Definition A.1.3: (*Intersection and union*). Let A_1, A_2 be subsets of a set Ω. Then A_1 *union* A_2, written as $A_1 \cup A_2$,is the set defined by

$$A_1 \cup A_2 = \{\omega : \omega \in A_1 \text{ or } \omega \in A_2 \text{ or both}\}.$$

Similarly, A_1 *intersection* A_2, written as $A_1 \cap A_2$, is the set defined by

$$A_1 \cap A_2 = \{\omega : \omega \in A_1 \text{ and } \omega \in A_2\}.$$

Example A.1.7: Let $\Omega \equiv \mathbb{N} \equiv \{1, 2, 3, \ldots\}$,

$$\begin{aligned} A_1 &= \{\omega : \omega = 3k \text{ for some } k \in \mathbb{N}\} \quad \text{and} \\ A_2 &= \{\omega : \omega \equiv 5k \text{ for some } k \in \mathbb{N}\}. \end{aligned}$$

Then $A_1 \cup A_2 = \{\omega : \omega$ is divisible by at least one of the two integers 3 and 5$\}$, $A_1 \cap A_2 = \{\omega : \omega$ is divisible by both 3 and 5$\}$.

Definition A.1.4: Let Ω and I be nonempty sets. Let $\{A_\alpha : \alpha \in I\}$ be a collection of subsets of Ω. Then I is called the *index set*.

The *union* of $\{A_\alpha : \alpha \in I\}$ is defined as

$$\bigcup_{\alpha \in I} A_\alpha \equiv \{\omega : \omega \in A_\alpha \text{ for some } \alpha \in I\}.$$

The *intersection* of $\{A_\alpha : \alpha \in I\}$ is defined as

$$\bigcap_{\alpha \in I} A_\alpha \equiv \{\omega : \omega \in A_\alpha \text{ for every } \alpha \in I\}.$$

Definition A.1.5: (*Complement of a set*). Let $A \subset \Omega$. Then the *complement* of the set A, written as A^c (or $\tilde{A}$), is defined by $A^c \equiv \{\omega : \omega \notin A\}$.

Example A.1.8: If $\Omega = \mathbb{N}$ and A is the set of all integers that are divisible by 2, then A^c is the set of all odd integers, i.e., $A^c = \{1, 3, 5, 7, \ldots\}$.

Proposition A.1.1: (*DeMorgan's law*). *For any $\{A_\alpha : a \in I\}$ of subsets of Ω, $(\cup_{\alpha \in I} A_\alpha)^c = \cap_{\alpha \in I} A_\alpha^c$, $(\cap_{\alpha \in I} A_\alpha)^c = \cup_{\alpha \in I} A_\alpha^c$.*

Proof: To show that two sets A and B are the same, it suffices to show that

$$\omega \in A \Rightarrow \omega \in B \quad \text{and} \quad \omega \in B \Rightarrow \omega \in A.$$

Let $\omega \in (\cup_{\alpha \in I} A_\alpha)^c$. Then $\omega \notin \cup_{\alpha \in I} A_\alpha$

$$\begin{aligned} \Rightarrow &\quad \omega \notin A_\alpha \text{ for any } \alpha \in I \\ \Rightarrow &\quad \omega \in A_\alpha^c \text{ for each } \alpha \in I \\ \Rightarrow &\quad \omega \in \bigcap_{\alpha \in I} A_\alpha^c. \end{aligned}$$

Thus $(\cup_{\alpha \in I} A_\alpha)^c \subset \cap_{\alpha \in I} A_\alpha^c$. The opposite inclusion and the second identity are similarly proved. □

Definition A.1.6: (*Product sets*). Let Ω_1 and Ω_2 be two nonempty sets. Then the *product set* of Ω_1 and Ω_2, denoted by $\Omega \equiv \Omega_1 \times \Omega_2$, consists of all ordered pairs (ω_1, ω_2) such that $\omega_1 \in \Omega_1, \omega_2 \in \Omega_2$.

Note that if $\Omega_1 = \Omega_2$ and $\omega_1 \neq \omega_2$, then the pair (ω_1, ω_2) is not the same as (ω_2, ω_1), i.e., the order is important.

Example A.1.9: $\Omega_1 = [0, 1]$, $\Omega_2 = [2, 3]$. Then $\Omega_1 \times \Omega_2 = \{(x, y) : 0 \leq x \leq 1,\ 2 \leq y \leq 3\}$.

Definition A.1.7: (*Finite products*). If $\Omega_i, i = 1, 2 \ldots, k$ are nonempty sets, then

$$\Omega = \Omega_1 \times \Omega_2 \times \ldots \times \Omega_k$$

is the set of all ordered k vectors $(\omega_1, \omega_2, \ldots, \omega_k)$ where $\omega_i \in \Omega_i$. If $\Omega_i = \Omega_1$ for all $1 \leq i \leq k$, then $\Omega_1 \times \Omega_2 \times \ldots \times \Omega_k$ is written as $\Omega_1^{(k)}$ or Ω_1^k.

Definition A.1.8: (*Infinite products*). Let $\{\Omega_\alpha : \alpha \in I\}$ be an infinite collection of nonempty sets. Then $\times_{\alpha \in I}\Omega_\alpha$, *the product set*, is defined as $\{f : f$ is a function defined on I such that for each α, $f(\alpha) \in \Omega_\alpha\}$. If $\Omega_\alpha = \Omega$ for all $\alpha \in I$, then $\times_{\alpha \in I}\Omega_\alpha$ is also written as Ω^I.

It is a *basic axiom of set theory*, known as the *axiom of choice* (A.C.), that this space is nonempty. That is, given an arbitrary collection of nonempty sets, it is possible to form a *parliament* with one representative from each set.

For a long time it was thought this should follow from the other axioms of set theory. But it is shown in Cohen (1966) that it is an independent axiom. That is, both the A.C. and its negation are consistent with the rest of the axioms of set theory.

There are several equivalent versions of A.C. These are Zorn's lemma, Hausdorff's maximality principle, the 'Principle of Well Ordering,' and Tukey's lemma. For a proof of these equivalences, see Hewitt and Stromberg (1965).

Definition A.1.9: (*Functions, countability and uncountability*). A *function* f is a correspondence between the elements of a set X and another set Y and is written as $f : X \to Y$. It satisfies the condition that for each x, there is a unique y in Y that corresponds to it and is denoted as

$$y = f(x).$$

The set X is called the *domain of* f and the set $f(X)$, defined as, $f(X) \equiv \{y :$ there exists x in X such that $f(x) = y\}$ is called the *range of* f. It is possible that many x's may correspond to the same y and also there may exist y in Y for which there is no x such that $f(x) = y$. If $f(X)$ is all of Y, then the map is called *onto*. If for each y in $f(X)$, there is a unique x in X such that $f(x) = y$, then f is called (1–1) or *one-to-one*. If f is one-to-one and onto, then X and Y are said to have the same *cardinality*.

Definition A.1.10: Let $f : X \to Y$ be (1–1) and onto. Then, for each y in Y, there is a unique element x in X such that $f(x) = y$. This x is denoted as $f^{-1}(y)$. Note that in this case, $g(y) \equiv f^{-1}(y)$ is a (1–1) onto map from Y to X and is called the *inverse* of f.

Example A.1.10: Let $X = \mathbb{N} \equiv \{1, 2, 3, \ldots\}$. Let $Y = \{n : n = 2k, k \in \mathbb{N}\}$ be the set of even integers. Then the map $f(x) = 2x$ is a (1–1) onto map from X to Y.

Example A.1.11: Let X be $\mathbb{N}$ and let $\mathbb{P}$ be the set of all prime numbers. Then X and $\mathbb{P}$ have the same cardinality.

Definition A.1.11: A set X is *finite* if there exists $n \in \mathbb{N}$ such that X and $Y \equiv \{1, 2, \ldots, n\}$ have the same *cardinality*, i.e., there exists a (1–1) onto map from Y to X. A set X is *countable* if X and $\mathbb{N}$ have the *same cardinality*, i.e., there exists a (1–1) onto map from $\mathbb{N}$ to X. A set X is *uncountable* if it is not *finite* or *countable*.

Example A.1.12: The set $\{0, 1, 2, \ldots, 9\}$ is finite, the set $\mathbb{N}^k$ ($k \in \mathbb{N}$) is countable and $\mathbb{N}^{\mathbb{N}}$ is uncountable (Problem A.6).

Definition A.1.12: Let Ω be a nonempty set. Then the power set of Ω, denoted by $\mathcal{P}(\Omega)$, is the collection of all subsets of Ω, i.e.,

$$\mathcal{P}(\Omega) \equiv \{A : A \subset \Omega\}.$$

Remark A.1.1: $\mathcal{P}(\mathbb{N})$ is an uncountable set (Problem A.5).

A.1.2 The principle of induction

The set $\mathbb{N}$ of natural numbers has the *well ordering property* that every nonempty subset A of $\mathbb{N}$ has a smallest element s such that (i) $s \in A$ and (ii) $a \in A \Rightarrow a \geq s$. This property is one of the basic postulates in the definition of $\mathbb{N}$. The *principle of induction* is a consequence of the well ordering property. It says the following:

Let $\{P(n) : n \in \mathbb{N}\}$ be a collection of propositions (or statements). Suppose that

(i) $P(1)$ is true.

(ii) For each $n \in \mathbb{N}$, $P(n)$ true $\Rightarrow P(n+1)$ true.

Then, $P(n)$ is true for all $n \in \mathbb{N}$.

See Problem A.9 for some examples.

A.1.3 Equivalence relations

Definition A.1.13:

(a) Let Ω be a nonempty set. Let G be a nonempty subset of $\Omega \times \Omega$. Write $x \sim y$ if $(x, y) \in G$ and call it a *relation* defined by G.

(b) A relation defined by G is an *equivalence relation* if

  (i) (*reflexive*) for all x in Ω, $x \sim x$, i.e., $(x, x) \in G$;

  (ii) (*symmetric*) $x \sim y \Rightarrow y \sim x$, i.e., $(x, y) \in G \Rightarrow (y, x) \in G$;

  (iii) (*transitive*) $x \sim y$, $y \sim z \Rightarrow x \sim z$, i.e., $(x, y) \in G$, $(y, z) \in G \Rightarrow (x, z) \in G$.

Example A.1.13: Let $\Omega = \mathbb{Z}$, the set of all integers, $G = \{(m,n) : m-n$ is divisible by $3\}$. Thus, $m \sim n$ if $(m-n)$ is a multiple of 3. It is easy to verify that this is an equivalence relation.

Definition A.1.14: (*Equivalence classes*). Let Ω be a nonempty set. Let G define an equivalence relation on Ω. For each x in Ω, the set $[x] \equiv \{y : x \sim y\}$ is called the *equivalence class generated by* x.

Proposition A.1.2: *Let $\mathcal{C}$ be the set of all equivalence classes in Ω generated by an equivalence relation defined by G. Then*

(i) $C_1, C_2 \in \mathcal{C} \Rightarrow C_1 = C_2$ *or* $C_1 \cap C_2 = \emptyset$.

(ii) $\bigcup_{C \in \mathcal{C}} C = \Omega$.

Proof:

(i) Suppose $C_1 \cap C_2 \neq \emptyset$. Then there exist x_1, x_2, y such that $C_1 = [x_1]$, $C_2 = [x_2]$ and $y \in C_1 \cap C_2$. This implies $x_1 \sim y$, $x_2 \sim y$. But by symmetry $y \sim x_2$ and this implies by transitivity that $x_1 \sim x_2$, i.e., $x_2 \in C_1$ implying $C_2 \subset C_1$. Similarly, $C_1 \subset C_2$, i.e., $C_1 = C_2$.

(ii) For each x in Ω, $(x,x) \in G$ and so $[x]$ is not empty and $x \in [x]$. $\square$

The above proposition says that every equivalence relation on Ω leads to a decomposition of Ω into equivalence classes that are disjoint and whose union is all of Ω. In the example given above, the set $\mathbb{Z}$ of all integers can be decomposed to three equivalence classes $\mathcal{C}_j \equiv \{n : n = 3m + j$ for some $m \in \mathbb{Z}\}$, $j = 0, 1, 2$.

A.2 Real numbers, continuity, differentiability, and integration

A.2.1 Real numbers

This section reviews the following: integers, rationals, real numbers; algebraic, order, and completeness axioms; Archimedean property, denseness of rationals.

There are at least two approaches to defining the real number system.

APPROACH 1. Start with the natural numbers $\mathbb{N}$, construct the set $\mathbb{Z}$ of all integers $(\mathbb{N} \cup \{0\} \cup (-\mathbb{N}))$, and next, the set $\mathbb{Q}$ of rationals and then the set $\mathbb{R}$ of real numbers either as the set of all Cauchy sequences of rationals or as Dedekind cuts. The step going from $\mathbb{Q}$ to $\mathbb{R}$ via Cauchy sequences is also available for completing any incomplete metric space (see Section A.4).

APPROACH 2. Define the set of real numbers $\mathbb{R}$ as a set that satisfies three sets of axioms. The first set is *algebraic* involving addition and multiplication. The second set is on *ordering* that, with the first, makes $\mathbb{R}$ an ordered field (see Royden (1988) for a definition). The third set is a single axiom known as the *completeness* axiom. Thus $\mathbb{R}$ is defined as a complete ordered field.

The *algebraic axioms* say that there are two binary operations known as addition $(+)$ and multiplication $(\cdot)$ that render $\mathbb{R}$ a field. See Royden (1988) for the nine axioms for this set.

The *order axiom* says that there is a set $\mathbb{P} \subset \mathbb{R}$, to be called positive numbers such that

(i) $x, y \in \mathbb{P} \Rightarrow x \cdot y \in \mathbb{P}$, $x + y \in \mathbb{P}$

(ii) $x \in \mathbb{P} \Rightarrow -x \notin \mathbb{P}$

(iii) $x \in \mathbb{R} \Rightarrow x = 0$ or $x \in \mathbb{P}$ or $-x \in \mathbb{P}$.

The set $\mathbb{Q}$ of rational numbers is an ordered field (i.e., it satisfies the algebraic and order axioms). But $\mathbb{Q}$ does not satisfy the completeness axiom (see below).

Given $\mathbb{P}$, one can define an order on $\mathbb{R}$ by defining $x < y$ (read x less than y) to mean $y - x \in \mathbb{P}$. Since for all x, y in $\mathbb{R}$, $(x - y)$ is either 0 or $(x - y) \in \mathbb{P}$ or $(y - x) \in \mathbb{P}$, it follows that for all x, y in $\mathbb{R}$, either $x = y$ or $x < y$ or $x > y$. This is called *total* or *linear order*.

Definition A.2.1: (*Upper and lower bounds*).

(a) Let $A \subset \mathbb{R}$. A real number M is an *upper bound* for A if $a \in A \Rightarrow a \leq M$ and m is a *lower bound* for A if $a \in A \Rightarrow a \geq m$.

(b) The *supremum* of a set A, denoted by *sup* A or the *least upper bound (l.u.b.)* of A, is defined by the following conditions:

  (i) $x \in A \Rightarrow x \leq \sup A$,

  (ii) $K < \sup A \Rightarrow$ there exists $x \in A$ such that $K < x$.

The completeness axiom says that if $A \subset \mathbb{R}$ has an upper bound M in $\mathbb{R}$, then there exists a $\tilde{M}$ in $\mathbb{R}$ such that $\tilde{M} = \sup A$.

That is, every set A that is bounded above in $\mathbb{R}$ has a l.u.b. in $\mathbb{R}$. The ordered field of rationals $\mathbb{Q}$ does not possess this property. One well-known example is the set

$$A = \{r : r \in \mathbb{Q}, r^2 < 2\}.$$

Then A is bounded above in $\mathbb{Q}$ but has no l.u.b. in $\mathbb{Q}$ (Problem A.11).

Next some consequences of the completeness axiom are discussed.

Proposition A.2.1: (*Axiom of Eudoxus and Archimedes (AOE)*). *For all x in $\mathbb{R}$, there exists a natural number n such that $n > x$.*

Proof: If $x \leq 1$, take $n = 2$. If $x > 1$, let $S_x \equiv \{k : k \in \mathbb{N}, k \leq x\}$. Then S_x is not empty and is bounded above. By the completeness axiom, there is a real number y that is the l.u.b. of S_x. Thus $y - \frac{1}{2}$ is not an upper bound for S_x and so there exists $k_0 \in S_x$ such that $y - \frac{1}{2} < k_0$. This implies that $(k_0 + 1) > y - \frac{1}{2} + 1 = y + \frac{1}{2} > y$ and so $(k_0 + 1) \notin S_x$. By the linear order in $\mathbb{R}$, $(k_0 + 1) > x$ and so $(k_0 + 1)$ is the desired integer. □

Corollary A.2.2: *For any x, $y \in \mathbb{R}$ with $x < y$, there is a r in $\mathbb{Q}$ such that $x < r < y$.*

Proof: Let $z = (y - x)^{-1}$. Then there is an integer k such that $0 < z < k$ (by AOE.) Again by AOE, there is a positive integer n such that $n > yk$. Let $S = \{n : n \in \mathbb{N}, n > yk\}$. Since $S \neq \emptyset$, it has a smallest element (by the well ordering property of $\mathbb{N}$) say, p. Then $p - 1 < yk < p$, i.e., $\frac{p-1}{k} < y < \frac{p}{k}$. Since $\frac{1}{k} < \frac{1}{z} = (y - x)$ and $\frac{p}{k} > y$, it follows that $\frac{p-1}{k} > x$. Now take $r = \frac{p-1}{k}$. □

Remark A.2.1: This property is often stated as: *The set $\mathbb{Q}$ of rationals is dense in the set $\mathbb{R}$ of real numbers.*

Definition A.2.2: The set $\bar{\mathbb{R}}$ of *extended real numbers* is the set consisting of $\mathbb{R}$ and two elements identified as $+\infty$ (plus infinity) and $-\infty$ (negative infinity) with the following definition of addition ($+$) and multiplication ($\cdot$). For any x in $\mathbb{R}$, $x + \infty = \infty$, $x - \infty = -\infty$, $x \cdot \infty = \infty$ if $x > 0$, $x \cdot \infty = -\infty$ if $x < 0$, $0 \cdot \infty = 0$, $\infty + \infty = \infty$, $-\infty - \infty = -\infty$, $\infty \cdot (\pm\infty) = \pm\infty$, $(-\infty) \cdot (\pm\infty) = \mp\infty$. But $\infty - \infty$ is not defined. The order property on $\bar{\mathbb{R}}$ is defined by extending that on $\mathbb{R}$ with the additional condition $x \in \mathbb{R} \Rightarrow -\infty < x < +\infty$. Finally, if $A \subset \mathbb{R}$ does not have an upper bound in $\mathbb{R}$, then $\sup A$ is defined as $+\infty$ and if $A \subset \mathbb{R}$ does not have a lower bound in $\mathbb{R}$, then $\inf A$ is defined as $-\infty$.

A.2.2 Sequences, series, limits, limsup, liminf

Definition A.2.3: Let $\{x_n\}_{n \geq 1}$ be a sequence of real numbers.

(i) For a real number a, $\lim_{n \to \infty} x_n = a$ if for every $\epsilon > 0$, there exists a positive integer N_ϵ such that $n \geq N_\epsilon \Rightarrow |x_n - a| < \epsilon$.

(ii) $\lim_{n \to \infty} x_n = \infty$ if for any K in $\mathbb{R}$, there exists an integer N_K such that $n \geq N_K \Rightarrow x_n > K$.

(iii) $\lim_{n \to \infty} x_n = -\infty$ if $\lim_{n \to \infty} (-x_n) = \infty$.

(iv) $\limsup_{n \to \infty} x_n \equiv \varlimsup_{n \to \infty} x_n = \inf_{n \geq 1} (\sup_{j \geq n} x_j)$.

(v) $\liminf_{n \to \infty} x_n \equiv \varliminf_{n \to \infty} x_n = \sup_{n \geq 1} (\inf_{j \geq n} x_j)$.

Definition A.2.4: (*Cauchy sequences*). A sequence $\{x_n\}_{n\geq 1} \subset \mathbb{R}$ is called a *Cauchy sequence* if for every $\varepsilon > 0$, there is N_ε such that $n, m \geq N_\varepsilon \Rightarrow |x_m - x_n| < \varepsilon$.

Proposition A.2.3: *If $\{x_n\}_{n\geq 1} \subset \mathbb{R}$ is convergent in $\mathbb{R}$ (i.e., $\lim_{n\to\infty} x_n = a$ exists in $\mathbb{R}$), then $\{x_n\}_{n\geq 1}$ is Cauchy. Conversely, if $\{x_n\}_{n\geq 1} \subset \mathbb{R}$ is Cauchy, then there exists an $a \in \mathbb{R}$ such that* $\lim_{n\to\infty} x_n = a$.

The proof is based on the use of the l.u.b. axiom (Problem A.14).

Definition A.2.5: Let $\{x_n\}_{n\geq 1}$ be a sequence of real numbers. For $n \geq 1$, $s_n \equiv \sum_{j=1}^{n} x_j$ is called the *nth partial sum* of the series $\sum_{j=1}^{\infty} x_j$. The series $\sum_{j=1}^{\infty} x_j$ is said to *converge* to s in $\mathbb{R}$ if $\lim_{n\to\infty} s_n = s$. If $\lim_{n\to\infty} s_n = \pm\infty$, then the series $\sum_{j=1}^{\infty} x_j$ is said to *diverge* to $\pm\infty$.

Note that if $x_j \geq 0$ for all j, then either $\lim_{n\to\infty} s_n = s \in \mathbb{R}$, or $\lim_{n\to\infty} s_n = \infty$.

Example A.2.1: (*Geometric series*). Fix $0 < r < 1$. Let $x_n = r^n$, $n \geq 0$. Then $s_n = 1 + r + \ldots + r^n = \frac{1-r^{n+1}}{1-r}$ and $\sum_{j=1}^{\infty} r^j$ converges to $s = \frac{1}{1-r}$.

Example A.2.2: Consider the series $\sum_{j=1}^{\infty} \frac{1}{j^p}$, $0 < p < \infty$. It can be shown that this converges for $p > 1$ and diverges to ∞ for $0 < p \leq 1$.

Definition A.2.6: The series $\sum_{j=1}^{\infty} x_j$ *converges absolutely* if the series $\sum_{j=1}^{\infty} |x_j|$ converges in $\mathbb{R}$.

There exist series $\sum_{j=1}^{\infty} x_j$ that converge but not absolutely. For example, $\sum_{j=1}^{\infty} \frac{(-1)^j}{j}$. For further material on convergence properties of series, such as tests for convergence, rates of convergence, etc., see Rudin (1976).

Definition A.2.7: (*Power series*). Let $\{a_n\}_{n\geq 0}$ be a sequence of real numbers. For $x \in \mathbb{R}$, the series $\sum_{n=0}^{\infty} a_n x^n$ is called a *power series*. If the series $\sum_{n=0}^{\infty} a_n x^n$ converges for all x in $B \subset \mathbb{R}$, the *power series* $\sum_{n=0}^{\infty} a_n x^n$ is said to be *convergent on* B.

Proposition A.2.4: *Let $\{a_n\}_{n\geq 0}$ be a sequence of real numbers. Let* $\rho = (\limsup_{n\to\infty} |a_n|^{\frac{1}{n}})^{-1}$. *Then*

(i) $|x| < \rho \Rightarrow \sum_{n=0}^{\infty} |a_n x^n|$ *converges.*

(ii) $|x| > \rho \Rightarrow \sum_{n=0}^{\infty} |a_n x^n|$ *diverges to* $+\infty$.

Proof of this is left as an exercise (Problem A.15).

Definition A.2.8: $\rho \equiv (\limsup_{n\to\infty} |a_n|^{\frac{1}{n}})^{-1}$ is called the *radius of convergence* of the power series $\sum_{n=0}^{\infty} a_n x^n$.

A.2.3 Continuity and differentiability

Definition A.2.9: Let $f : A \to \mathbb{R}$, $A \subset \mathbb{R}$. Then

(a) f is *continuous at* x_0 in A if for every $\epsilon > 0$, there exists a $\delta > 0$ such that $x \in A$, $|x - x_0| < \delta$, implies $|f(x) - f(x_0)| < \epsilon$. (Here, δ may depend on ϵ and x_0.)

(b) f is *continuous on* $B \subset A$ if it is continuous at every x_0 in B.

(c) f is *uniformly continuous* on $B \subset A$ if for every $\epsilon > 0$, there exists a $\delta_\epsilon > 0$ such that $\sup\{|f(x) - f(y)| : x, y \in B, |x - y| < \delta_\epsilon\} < \epsilon$.

Some properties of continuous functions are listed below.

Proposition A.2.5:

(i) (*Sums, products, and ratios of continuous functions*). *Let $f, g : A \to \mathbb{R}$, $A \subset \mathbb{R}$. Let f and g be continuous on $B \subset A$. Then*

- *(a) $f + g$, $f - g$, $\alpha \cdot f$ for any $\alpha \in \mathbb{R}$ are all continuous on B.*
- *(b) $f(x)/g(x)$ is continuous at x_0 in B, provided $g(x_0) \neq 0$.*

(ii) (*Continuous functions on a closed bounded interval*). *Let f be continuous on a closed and bounded interval $[a, b]$. Then*

- *(a) f is bounded, i.e., $\sup\{|f(x)| : a \leq x \leq b\} < \infty$,*
- *(b) it achieves its maximum and minimum, i.e., there exist x_0, y_0 in $[a, b]$ such that $f(x_0) \geq f(x) \geq f(y_0)$ for all x in $[a, b]$ and f attains all values in $[f(y_0), f(x_0)]$, i.e., for all $\ell \in [f(y_0), f(x_0)]$, there exists $z \in [a, b]$ such that $f(z) = \ell$. Thus, f maps bounded closed intervals onto bounded closed intervals.*
- *(c) f is uniformly continuous on $[a, b]$.*

(iii) (*Composition of functions*). *Let $f : A \to \mathbb{R}$, $g : B \to \mathbb{R}$ be continuous on A and B, respectively. Let $f(A) \subset B$, i.e., for any x in A, $f(x) \in B$. Let $h(x) = g(f(x))$ for x in A. Then $h : A \to \mathbb{R}$ is continuous.*

(iv) (*Uniform limits of continuous functions*). *Let $\{f_n\}_{n \geq 1}$, be a sequence of functions continuous on A to $\mathbb{R}$, $A \subset \mathbb{R}$. If $\sup\{|f_n(x) - f(x)| : x \in A\} \to 0$ as $n \to \infty$ for some $f : A \to \mathbb{R}$, i.e., f_n converges to f uniformly on A, then f is continuous on A.*

Remark A.2.2: The function $f(x) \equiv x$ is clearly continuous on $\mathbb{R}$. Now by Proposition A.2.5 (i) and (iv), it follows that all polynomials are continuous on $\mathbb{R}$, and hence, so are their uniform limits. *Weierstrass' approximation theorem* is a sort of converse to this. That is, every continuous function on a closed and bounded interval is the uniform limit of polynomials. More precisely, one has the following:

Theorem A.2.6: *Let $f : [a,b] \to \mathbb{R}$ be continuous. Then for any $\epsilon > 0$ there is a polynomial $p(x) = \sum_0^n a_j x^j$, $a_j \in \mathbb{R}$, $j = 0,1,2,\ldots,n$ such that* $\sup\{|f(x) - p(x)| : x \in [a,b]\} < \epsilon$.

It should be noted that a power series $A(x) \equiv \sum_0^\infty a_n x^n$ is the uniform limits of polynomials on $[-\lambda, \lambda]$ for any $0 < \lambda < \rho \equiv \left(\overline{\lim}_{n\to\infty} |a_n|^{1/n}\right)^{-1}$ and hence is continuous on $(-\rho, \rho)$.

Definition A.2.10: Let $f : (a,b) \to \mathbb{R}$, $(a,b) \subset \mathbb{R}$. The function f is said to be *differentiable at* $x_0 \in (a,b)$ if

$$\lim_{h\to 0} \frac{f(x_0+h) - f(x_0)}{h} \equiv f'(x_0) \quad \text{exists in} \quad \mathbb{R}.$$

A function is differentiable in (a,b) if it is differentiable at each x in (a,b).

Some important consequences of differentiability are listed below.

Proposition A.2.7: *Let f, $g : (a,b) \to \mathbb{R}$, $(a,b) \subset \mathbb{R}$. Then*

(i) *f differentiable at x_0 in (a,b) implies f is continuous at x_0.*

(ii) *(Mean value theorem). f differentiable on (a,b), f continuous on $[a,b]$ implies that for some $a < c < b$, $f(b) - f(a) = (b-a)f'(c)$.*

(iii) *(Maxima and minima). f differentiable at x_0 and for some $\delta > 0$, $f(x) \le f(x_0)$ for all $x \in (x_0 - \delta, x_0 + \delta)$ implies that $f'(x_0) = 0$.*

(iv) *(Sums, products and ratios). f, g differentiable at x_0 implies that for any α, β in $\mathbb{R}$, $(\alpha f + \beta g)$, $f - g$ are differentiable at x_0 with*

$$\begin{aligned} (\alpha f + \beta g)'(x_0) &= \alpha f'(x_0) + \beta g'(x_0), \\ (fg)'(x_0) &= f'(x_0)g(x_0) + f(x_0)g'(x_0), \end{aligned}$$

and if $g'(x_0) \neq 0$, then f/g is differentiable at x_0 with

$$(f/g)'(x_0) = \frac{f'(x_0)g(x_0) - f(x_0)g'(x_0)}{(g(x_0))^2}.$$

(v) *(Chain rule). If f is differentiable at x_0 and g is differentiable at $f(x_0)$, then $h(x) \equiv g(f(x))$ is differentiable at x_0 with $h'(x_0) = g'(f(x_0))f'(x_0)$.*

(vi) *(Differentiability of power series). Let $A(x) \equiv \sum_{n=0}^\infty a_n x^n$ be a power series with radius of convergence $\rho \equiv \left(\overline{\lim}_{n\to\infty} |a_n|^{1/n}\right)^{-1} > 0$. Then $A(\cdot)$ is differentiable infinitely many times on $(-\rho, \rho)$ and for x in $(-\rho, \rho)$,*

$$\frac{d^k A(x)}{dx^k} = \sum_{n=k}^\infty n(n-1)\cdots(n-k+1)x^{n-k}, \ k \ge 1.$$

Remark A.2.3: It should be noted that the converse to (a) in the above proposition does not hold. For example, the function $f(x) = |x|$ is continuous at $x_0 = 0$ but is not differentiable at x_0. Indeed, Weierstrass showed that there exists a function $f : [0,1] \to \mathbb{R}$ such that it is continuous on $[0,1]$ but is not differentiable at any x in $(0,1)$.

Also note that the mean value theorem implies that if $f'(\cdot) \geq 0$ on (a,b), then f is nondecreasing on (a,b).

Definition A.2.11: (*Taylor series*). Let f be a map from $I \equiv (a-\eta, a+\eta)$ to $\mathbb{R}$ for some $a \in \mathbb{R}$, $\eta > 0$. Suppose f is n times differentiable in I, for each $n \geq 1$. Let $a_n = \frac{f^{(n)}(a)}{n!}$. Then power series $\sum_{n=0}^{\infty} a_n (x-a)^n = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!}(x-a)^n$ is called the *Taylor series* of f at a.

Remark A.2.4: Let f be as in Definition A.2.11. *Taylor's remainder theorem* says that for any x in I and any $n \geq 1$, if f is $(n+1)$ times differentiable in I, then

$$\left| f(x) - \sum_{j=0}^{n} a_j x^j \right| \leq \frac{|f^{(n+1)}(y_n)|}{(n+1)!}$$

for some y_n in I.

Thus, if for some $\epsilon > 0$, $\sup_{|y-a|<\varepsilon} |f^{(k)}(y)| \equiv \lambda_k$ satisfies $\frac{\lambda_k}{k!} \to 0$ as $k \to \infty$, then the Taylor series satisfies

$$\sup_{|x-a|<\varepsilon} \left| \sum_{j=0}^{n} (x-a)^j \frac{f^{(j)}(a)}{j!} - f(x) \right| \to 0 \quad \text{as} \quad n \to \infty.$$

A.2.4 Riemann integration

Let $f : [a,b] \to \mathbb{R}$, $[a,b] \subset \mathbb{R}$. Let $\sup\{|f(x)| : a \leq x \leq b\} < \infty$. For any partition $P \equiv \{a = x_0 < x_1 < x_2 < x_k = b\}$, let

$$\begin{aligned} U(P,f) &\equiv \sum_{i=1}^{n} M_i(P)\Delta_i(P) \quad \text{and} \\ L(P,f) &\equiv \sum_{i=1}^{n} m_i(P)\Delta_i(P) \end{aligned}$$

where

$$\begin{aligned} M_i(P) &= \sup\{f(x) : x_i \leq x \leq x_i\}, \\ m_i(P) &= \inf\{f(x) : x_{i-1} \leq x \leq x_i\}, \\ \Delta_i(P) &= (x_i - x_{i-1}) \end{aligned}$$

for $i = 1, 2, \ldots, k$.

Definition A.2.12: The upper and lower Riemann integrals of f over $[a, b]$, denoted by $\overline{\int}_a^b f(x)dx$ and $\underline{\int}_a^b f(x)dx$, respectively defined as

$$\begin{aligned} \overline{\int}_a^b f(x)dx &\equiv \inf\{U(P,f) : P \text{ a partition of } [a,b]\} \\ \underline{\int}_a^b f(x)dx &\equiv \sup\{L(P,f) : P \text{ a partition of } [a,b]\}. \end{aligned}$$

Definition A.2.13: Let $f : [a, b] \to \mathbb{R}$, $[a, b] \subset \mathbb{R}$ and let $\sup\{|f(x)| : a \leq x \leq b\} < \infty$. The function f is Riemann integrable on $[a, b]$ if

$$\overline{\int}_a^b f(x)dx = \underline{\int}_a^b f(x)dx$$

and the common value is denoted as $\int_a^b f(x)dx$.

The following are some important results on Riemann integration.

Proposition A.2.8:

(i) *Let $f : [a, b] \to \mathbb{R}$, $[a, b] \subset \mathbb{R}$ be continuous. Then f is Riemann integrable and $\int_a^b f(x)dx = \lim_{n\to\infty} \frac{1}{k_n} \sum_{i=0}^{k_n} f(x_{ni})\Delta_{ni}$ where $\{P_n \equiv \{x_{ni} : 1 \leq i \leq k_n\}\}$ is a sequence of partitions of $[a, b]$ such that $\Delta_n \equiv \{\max(x_{ni} - x_{n,i-1}) : 1 \leq i \leq k_n\} \to 0$ as $n \to \infty$.*

(ii) *If f, g are Riemann integrable on $[a, b]$, then $\alpha f + \beta g$, $\alpha, \beta \in \mathbb{R}$ and fg are Riemann integrable on $[a, b]$ and*

$$\int_a^b (\alpha f + \beta g)(x)dx = \alpha \int_a^b f(x)dx + \beta \int_a^b g(x)dx.$$

(iii) *Let f be Riemann integrable on $[a, b]$ and $[b, c]$, $-\infty < a < b < c < \infty$. Then f is Riemann integrable on $[a, c]$ and*

$$\int_a^c f(x)dx = \int_a^b f(x)dx + \int_b^c f(x)dx.$$

(iv) *(Fundamental theorem of Riemann integration: Part I). Let f be Riemann integrable on $[a, b]$. Then it is Riemann integrable on $[a, c]$ for all $a \leq c \leq b$. Let $F(x) \equiv \int_a^x f(u)du$, $a \leq x \leq b$. Then*

- *(a)* *$F(\cdot)$ is continuous on $[a, b]$.*
- *(b)* *If $f(\cdot)$ is continuous at $x_0 \in (a, b)$, then $F(\cdot)$ is differentiable at x_0 and $F'(x_0) = f(x_0)$.*

(v) (*Fundamental theorem: Part II*). *If $F : [a,b] \to \mathbb{R}$ is differentiable on (a,b), $f : [a,b] \to \mathbb{R}$ is continuous on $[a,b]$ and $F'(x) = f(x)$ for all $a < x < b$, then $F(x) = F(a) + \int_a^c f(x)dx$, $a \le c \le b$.*

(vi) (*Integration by parts*). *If f and g are continuous and differentiable on (a,b) with f' and g' continuous on (a,b), then*

$$\begin{aligned}\int_c^d f(x)g'(x)dx + \int_c^d f'(x)g(x)dx \\ = \quad f(d)g(d) - f(c)g(c) \quad \text{for all} \quad a < c < d < b.\end{aligned}$$

(vii) (*Interchange of limits and integration*). *Let f_n, f be continuous on $[a,b]$ to $\mathbb{R}$. Let f_n converge to f uniformly on $[a,b]$. Then*

$$F_n(c) \equiv \int_a^c f_n(x)dx \to F(c) \equiv \int_a^c f(x)dx$$

uniformly on $[a,b]$.

A.3 Complex numbers, exponential and trigonometric functions

Definition A.3.1: The set $\mathbb{C}$ of complex numbers is defined as the set $\mathbb{R} \times \mathbb{R}$ of all ordered pairs of real numbers endowed with addition and multiplication as follows:

$$\begin{aligned}(a_1,b_1) + (a_2,b_2) &= (a_1+a_2, b_1+b_2) \\ (a_1,b_1)\cdot(a_2,b_2) &= (a_1a_2 - b_1b_2, a_1b_2 + a_2b_1).\end{aligned} \tag{3.1}$$

It can be verified that $\mathbb{C}$ satisfies the field axioms Royden (1988). By defining ι to be the element (0,1), one can write the elements of $\mathbb{C}$ in the form $(a,b) = a + \iota b$ and do addition and multiplication with the rule of replacing ι^2 by -1. Clearly, the set $\mathbb{R}$ of real numbers can be identified with the set $\{(a,0) : a \in \mathbb{R}\}$. The set $\{(0,b) : b \in \mathbb{R}\}$ is called the set of *purely imaginary numbers*.

Definition A.3.2: For any complex number $z = (a,b)$ in $\mathbb{C}$,

$$\begin{aligned}\mathrm{Re}(z),&\quad \text{called the } \textit{real part of } z \text{ is } a,\\ \mathrm{Im}(z),&\quad \text{called the } \textit{imaginary part of } z \text{ is } b,\\ \bar{z},&\quad \text{called the } \textit{complex conjugate of } z \text{ is } \bar{z} = a - \iota b\\ |z|,&\quad \text{called the } \textit{absolute value of } z \text{ is } |z| = \sqrt{a^2+b^2}\ .\end{aligned} \tag{3.2}$$

Clearly, $z\bar{z} = \bar{z}z = |z|^2$ and any $z \neq 0$ can be written as $z = |z|\omega$ where $|\omega| = 1$. A set $A \subset \mathbb{C}$ is *bounded* if $\sup\{|z| : z \in A\} < \infty$. If $d(z_1, z_2) \equiv |z_1 - z_2|$, then $(\mathbb{C}, d)$ is a *complete separable metric space* (see Section A.4). Clearly, $z_n = (a_n, b_n)$ converges to $z = (a, b)$ in this metric iff $\mathrm{Re}(z_n) = a_n \to \mathrm{Re}(z) = a$ and $\mathrm{Im}(z_n) = b_n \to \mathrm{Im}(z) = b$.

Definition A.3.3: The *exponential function* is a map from $\mathbb{C}$ to $\mathbb{C}$ defined by

$$\exp(z) \equiv \sum_{n=0}^{\infty} \frac{z^n}{n!} \tag{3.3}$$

where the right side is defined as the limit of the partial sum sequence $\{\sum_{n=0}^{m} \frac{z^n}{n!}\}_{m \geq 0}$, which exists since $\sum_{n=0}^{\infty} \frac{|z|^n}{n!} < \infty$ for each $z \in \mathbb{C}$. In fact, the convergence of the partial sum sequence is uniform on every bounded subset A of $\mathbb{C}$. This in turn implies that the exponential function is continuous on $\mathbb{C}$.

Notation: $\exp(z)$ will also be written as e^z.

Definition A.3.4: The number $e^1 \equiv \sum_{n=0}^{\infty} \frac{1}{n!}$ will be called e. It is not difficult to show that $e = \lim_{n\to\infty}(1 + \frac{1}{n})^n$. By definition of e^z, $e^0 = 1$.

Definition A.3.5: The *cosine* and *sine* functions from $\mathbb{R} \to \mathbb{R}$ are defined by

$$\begin{aligned}
\cos t &\equiv \mathrm{Re}(e^{\iota t}) = \sum_{k=0}^{\infty} (-1)^k \frac{t^{2k}}{(2k)!} \\
\sin t &\equiv \mathrm{Im}(e^{\iota t}) = \sum_{k=0}^{\infty} (-1)^k \frac{t^{2k+1}}{(2k+1)!}.
\end{aligned}$$

Thus, one gets Euler's formula $e^{\iota t} = \cos t + \iota \sin t$, for all $t \in \mathbb{R}$.

The following are some important and useful properties of the exponential and the cosine and sine (also called *trigonometric*) functions.

Theorem A.3.1:

(i) For all z_1, z_2 in $\mathbb{C}$

$$e^{z_1} e^{z_2} = e^{z_1 + z_2} \quad \textit{and} \quad e^z \neq 0 \quad \textit{for any} \quad z.$$

(ii) e^z is differentiable for all z (i.e., $(e^z)' \equiv \lim_{h\to 0} \frac{e^{z+h} - e^z}{h}$ exists) and $(e^z)' = e^z$.

(iii) The function $e^x : \mathbb{R} \to \mathbb{R}_+$ is strictly increasing with $e^x \uparrow \infty$ as $x \uparrow \infty$ and $\downarrow 0$ as $x \downarrow -\infty$.

(iv) For t in $\mathbb{R}$, $(\cos t)^2 + (\sin t)^2 = 1$, $(\cos t)' = -\sin t$, $(\sin t)' = \cos t$.

(v) *There is a smallest positive number, called* π, *such that* $e^{\iota\pi/2} = \iota$.

(vi) e^z *is a periodic function such that* $e^z = e^{z+\iota 2\pi}$.

Proof:

(i) Since $\sum_{n=0}^{\infty} \frac{|z|^n}{n!}$ converges for all $z \in \mathbb{C}$,

$$\begin{aligned} e^{z_1} \cdot e^{z_2} &= \lim_{m\to\infty} \left(\sum_{n=0}^{m} \frac{z_1^n}{n!}\right)\left(\sum_{n=0}^{m} \frac{z_2^n}{n!}\right) \\ &= \lim_{m\to\infty} \sum_{0\le r,s\le m} \frac{z_1^r z_2^s}{r!s!} \\ &= \lim_{m\to\infty} \sum_{k=0}^{2m} \frac{1}{k!} \sum_{r=0}^{k} \frac{k!}{r!(k-r)!} z_1^r z_2^{k-r} \\ &= \lim_{m\to\infty} \sum_{k=0}^{2m} \frac{(z_1+z_2)^k}{k!} = e^{z_1+z_2}. \end{aligned}$$

Since $e^z \cdot e^{-z} = e^0 = 1$, $e^z \neq 0$ for any $z \in \mathbb{C}$.

(ii) Fix $z \in \mathbb{C}$. For any $h \in \mathbb{C}$, $h \neq 0$, by (i),

$$\frac{e^{z+h} - e^z}{h} = e^z \frac{e^h - 1}{h}.$$

But

$$\begin{aligned} \left|\frac{e^h - 1}{h} - 1\right| &\le \left(\sum_{k=2}^{\infty} \frac{|h|^k}{k!}\right) \frac{1}{|h|} \\ &\le |h| \sum_{k=0}^{\infty} \frac{1}{k!} \quad \text{if} \quad |h| \le 1. \end{aligned}$$

Thus

$$\lim_{h\to 0} \left(\frac{e^h - 1}{h} - 1\right) = 0$$

and

$$\lim_{h\to 0} \left(\frac{e^{z+h} - e^z}{h}\right) = e^z, \quad \text{i.e. (ii) holds.}$$

(iii) That the map $t \to e^t$ is strictly increasing on $[0, \infty)$ and that $e^t \uparrow \infty$ as $t \uparrow \infty$ is clear from the definition of e^z. Since $e^{-t}e^t = e^0 = 1$, $e^{-t} = \frac{1}{e^t}$ for all $t \in \mathbb{R}$, so that e^t is strictly increasing on all of $\mathbb{R}$ and $e^t \downarrow 0$ as $t \downarrow -\infty$.

(iv) From the definition of e^z, it follows that $\overline{e^z} = e^{\bar{z}}$ and, in particular, for t real $\overline{e^{\iota t}} = e^{\overline{\iota t}} = e^{-\iota t}$ and hence

$$|e^{\iota t}|^2 = \overline{e^{\iota t}} e^{\iota t} = e^{-\iota t + \iota t} = e^0 = 1.$$

Thus, for all t real $|e^{\iota t}| = 1$ and since $e^{\iota t} = \cos t + \iota \sin t$, it follows that

$$|e^{\iota t}|^2 = (\cos t)^2 + (\sin t)^2 = 1.$$

Also from (ii), for $t \in \mathbb{R}$

$$(e^{\iota t})' = (\cos t)' + \iota(\sin t)' = \iota e^{\iota t} = -\sin t + \iota \cos t$$

yielding $(\cos t)' = -\sin t$, $(\sin t)' = \cos t$ proving (iv).

(v) By definition

$$\cos 2 = \sum_{k=0}^{\infty} (-1)^k \frac{2^{2k}}{(2k)!} .$$

Now $a_k \equiv \frac{2^{2k}}{(2k)!}$ satisfies $a_{k+1} < a_k$ for $k = 0, 1, 2, \ldots$ and hence $\sum_{k \geq 3} (-1)^k a_k = -\{(a_3 - a_4) + (a_5 - a_6) + \cdots\} < 0$. Thus,

$$\cos 2 < 1 - \frac{2^2}{2!} + \frac{2^4}{4!} = -\frac{1}{3} < 0.$$

Since $\cos t$ is a continuous function on $\mathbb{R}$ with $\cos 0 = 1$ and $\cos 2 < 0$, there exists a smallest $t_0 > 0$ such that $\cos t_0 = 0$, defined by $t_0 = \inf\{t : t > 0, \cos t = 0\}$. Set $\pi = 2t_0$. Since $\cos \frac{\pi}{2} = 0$, (iv) implies $\sin \frac{\pi}{2} = 1$ and hence that $e^{\iota \frac{\pi}{2}} = \iota$.

(vi) Clearly, $e^{\iota \frac{\pi}{2}} = \iota$ implies that $e^{\iota \pi} = -1$ and $e^{\iota 2\pi} = 1$ and $e^{\iota 2\pi k} = 1$ for all integers k. Since $e^{\iota 2\pi} = 1$, it follows that $e^z = e^{z + \iota 2\pi}$ for all $z \in \mathbb{C}$,

$\square$

It is now possible to prove various results involving π that one learns in calculus from the above definition. For example, that the arc length of the *unit circle* $\{z : |z| = 1\}$ is 2π and that $\int_{-\infty}^{\infty} \frac{1}{1+x^2} dx = \pi$, etc. (Problems A.19 and A.20).

The following assertions about e^z can be proved with some more effort.

Theorem A.3.2:

(i) $e^z = 1$ *iff* $z = 2\pi\iota k$ *for some integer* k.

(ii) The map $t \to e^{\iota t}$ *from* $\mathbb{R}$ *is* onto *the unit circle.*

(iii) For any $\omega \in \mathbb{C}$, $\omega \neq 0$ *there is a* $z \in \mathbb{C}$ *such that* $\omega = e^z$.

For a proof of this theorem as well as more details on Theorem A.3.1, see Rudin (1987).

Theorem A.3.3: (*Orthogonality of* $\{e^{\iota 2\pi nt}\}_{n\in\mathbb{Z}}$). *For any* $n \in \mathbb{Z}$, $\int_0^1 e^{\iota 2\pi nt}dt = 0$ *if* $n \neq 0$ *and 1 if* $n = 0$.

Proof: Since $(e^{\iota t})' = \iota e^{\iota t}$

$$(e^{\iota 2\pi nt})' = \iota 2\pi n\, e^{\iota 2\pi nt},\ n \in \mathbb{Z}$$

and so for $n \neq 0$,

$$\begin{aligned}\int_0^1 e^{\iota 2\pi nt}dt &= \frac{1}{\iota 2\pi n}\int_0^1 (e^{\iota 2\pi nt})' dt \\ &= \frac{1}{\iota 2\pi n}(e^{\iota 2\pi n} - 1) = 0.\end{aligned}$$

Corollary A.3.4: *The family* $\{\cos 2\pi nt : n = 0, 1, 2, \ldots\} \cup \{\sin 2\pi nt : n = 1, 2, \ldots\}$ *are orthogonal in* $L^2[0,1]$ *(Problem A.22), i.e., for any two* f, g *in this family,* $\int_0^1 f(x)g(x)dx = 0$ *for* $f \neq g$.

A.4 Metric spaces

A.4.1 Basic definitions

This section reviews the following: metric spaces, Cauchy sequences, completeness, functions, continuity, compactness, convergence of sequences functions, and uniform convergence.

Definition A.4.1: Let $\mathbb{S}$ be a nonempty set. Let $d : \mathbb{S} \times \mathbb{S} \to \mathbb{R}_+ = [0, \infty)$ be such that

(i) $d(x, y) = d(y, x)$ for any x, y in $\mathbb{S}$.

(ii) $d(x, z) \leq d(x, y) + d(y, z)$ for any x, y, z in $\mathbb{S}$.

(iii) $d(x, y) = 0$ iff $x = y$.

Such a d is called a *metric* on $\mathbb{S}$ and the pair $(\mathbb{S}, d)$ a *metric space*. Property (ii) is called the *triangle inequality*.

Example A.4.1: Let $\mathbb{R}^k \equiv \{(x_1, \ldots, x_k) : x_i \in \mathbb{R},\ 1 \leq i \leq k\}$ be the k-dimensional Euclidean space. For $1 \leq p < \infty$ and $x = (x_1, \ldots, x_k),\ y = (y_1, y_2, \ldots, y_k) \in \mathbb{R}^k$, let

$$d_p(x, y) = \left(\sum_{i=1}^{k} |x_i - y_i|^p \right)^{\frac{1}{p}},$$

and $d_\infty(x, y) = \max\{|x_i - y_i| : 1 \leq i \leq k\}$.

It can be shown that $d_p(\cdot, \cdot)$ is a metric on $\mathbb{R}$ for all $1 \leq p \leq \infty$ (Problem A.24).

Definition A.4.2: A sequence $\{x_n\}_{n\geq 1}$ in a metric space $(\mathbb{S}, d)$ *converges* to an x in $\mathbb{S}$ if for every $\varepsilon > 0$, there is a N_ε such that $n \geq N_\varepsilon \Rightarrow d(x_n, x) < \varepsilon$ and is written as $\lim_{n\to\infty} x_n = x$.

Definition A.4.3: A sequence $\{x_n\}_{n\geq 1}$ in a metric space $(\mathbb{S}, d)$ is *Cauchy* if for all $\varepsilon > 0$, there exists N_ε such that $n, m \geq N_\varepsilon \Rightarrow d(x_n, x_m) < \varepsilon$.

Definition A.4.4: A metric space $(\mathbb{S}, d)$ is *complete* if every Cauchy sequence $\{x_n\}_{n\geq 1}$ converges to some x in $\mathbb{S}$.

Example A.4.2:

(a) Let $\mathbb{S} = \mathbb{Q}$, the set of rationals and $d(x, y) \equiv |x - y|$. Then $(\mathbb{Q}, d)$ is a metric space that is *not* complete.

(b) Let $\mathbb{S} = \mathbb{R}$ and $d(x, y) = |x - y|$. Then $(\mathbb{R}, d)$ is *complete* (cf. Proposition A.2.3).

(c) Let $\mathbb{S} = \mathbb{R}^k$. Then $(\mathbb{R}^k, d_p)$ is complete for every $1 \leq p \leq \infty$, where d_p is as in Example A.4.1.

Remark A.4.1: (*Completion of an incomplete metric space*). Let $(\mathbb{S}, d)$ be a metric space. Let $\tilde{\mathbb{S}}$ be the set of all Cauchy sequences in $\mathbb{S}$. Identify each x in $\mathbb{S}$ with the Cauchy sequence $\{x_n = x\}_{n\geq 1}$. Define a function from $\tilde{\mathbb{S}} \times \tilde{\mathbb{S}}$ to $\mathbb{R}_+$ by

$$\tilde{d}(\{x_n\}_{n\geq 1}, \{y_n\}_{n\geq 1}) = \limsup_{n\to\infty} \ d(x_n, y_n).$$

It is easy to verify that $\tilde{d}$ is symmetric and satisfies the triangle inequality. Define $s_1 = \{x_n\}_{n\geq 1}$ and $s_2 = \{y_n\}_{n\geq 1}$ to be *equivalent* (write $\{x_n\} \sim \{y_n\}$) if $\tilde{d}(s_1, s_2) = 0$. Let $\bar{\mathbb{S}}$ be the set of all equivalence classes in $\tilde{\mathbb{S}}$ and define $\bar{d}(c_1, c_2) \equiv \tilde{d}(s_1, s_2)$, where c_1, c_2 are equivalence classes and s_1, s_2 are arbitrary elements of c_1 and c_2, respectively.

It can now be verified that $(\bar{\mathbb{S}}, \bar{d})$ is a complete metric space and $(\mathbb{S}, d)$ is embedded in $(\bar{\mathbb{S}}, \bar{d})$ by identifying each x in $\mathbb{S}$ with the equivalence class containing the sequence $\{x_n = x\}_{n\geq 1}$.

Definition A.4.5: A metric space $(\mathbb{S}, d)$ is *separable* if there exists a subset $D \subset \mathbb{S}$ that is countable and *dense in* $\mathbb{S}$, i.e., for each x in $\mathbb{S}$ and $\varepsilon > 0$, there is a y in D such that $d(x, y) < \varepsilon$.

Example A.4.3: By the Archimedean property, $\mathbb{Q}$ is dense in $\mathbb{R}$. Similarly $\mathbb{Q}^k$, the set of all k vectors with components from $\mathbb{Q}$, is dense in $\mathbb{R}^k$.

Definition A.4.6: A metric space $(\mathbb{S}, d)$ is called *Polish* if it is complete and separable.

Example A.4.4: $(\mathbb{R}^k, d_p)$ in Example A.4.2 is Polish.

A.4.2 Continuous functions

Let $(\mathbb{S}, d)$ and $(\mathbb{T}, \rho)$ be two metric spaces. Let $f : \mathbb{S} \to \mathbb{T}$ be a map from $\mathbb{S}$ to $\mathbb{T}$.

Definition A.4.7:

(a) f is *continuous at* p in $\mathbb{S}$ if for each $\varepsilon > 0$, there exists $\delta > 0$ such that $d(x, p) < \delta \Rightarrow \rho(f(x), f(p)) < \varepsilon$. (Here the δ may depend on ε and p.)

(b) f is *continuous on a set* $B \subset \mathbb{S}$ if it is continuous at every $p \in B$.

(c) f is *uniformly continuous* on B if for each $\varepsilon > 0$, there exists $\delta > 0$ such that for each pair x, y in $\mathbb{S}$, $d(x, y) < \delta \Rightarrow \rho(f(x), f(y)) < \varepsilon$.

Definition A.4.8: Let $(\mathbb{S}, d)$ be a metric space.

(a) A set $O \subset (\mathbb{S}, d)$ is *open* if $x \in O \Rightarrow$ there exists $\delta > 0$ such that $d(x, y) < \delta \Rightarrow y \in O$. That is, at every point x in O, an *open ball* $B_x(\delta) \equiv \{y : d(x, y) < \delta\}$ of positive radius δ is a subset of O.

(b) A set $C \subset (\mathbb{S}, d)$ is *closed* if C^c is open.

Theorem A.4.1: *Let* $(\mathbb{S}, d)$ *and* $(\mathbb{T}, \rho)$ *be metric spaces. A map* $f : \mathbb{S} \to \mathbb{T}$ *in continuous on* $\mathbb{S}$ *iff for each* O *open in* $\mathbb{T}$, $f^{-1}(O)$ *is open in* $\mathbb{S}$.

Proof is left as an exercise (Problem A.28).

A.4.3 Compactness

Definition A.4.9: A collection of open sets $\{O_\alpha : \alpha \in I\}$ is an *open cover* for a set $B \subset (\mathbb{S}, d)$ if for each $x \in B$, there exists $\alpha \in I$ such that $x \in O_\alpha$.

Example A.4.5: Let $B = (0, 1)$. Then the collection $\{(\alpha - \frac{\alpha}{2}, \alpha + \frac{(1-\alpha)}{2}) : \alpha \in \mathbb{Q} \cap (0, 1)\}$ is an open cover for B.

Definition A.4.10: Let $(\mathbb{S}, d)$ be a metric space. A set $K \subset \mathbb{S}$ is called *compact* if given any open cover $\{O_\alpha : \alpha \in I\}$ for K, there exists a finite subcollection $\{O_{\alpha_i} : \alpha_i \in I, i = 1, 2, \ldots, n, n < \infty\}$ that is an open cover for K.

Example A.4.6: The set $B = (0, 1)$ is not *compact* as the open cover in the above Example A.3.4 does not admit a finite subcover.

The next result is the well-known *Heine-Borel theorem.*

Theorem A.4.2:

(i) For any $-\infty < a < b < \infty$, the closed interval $[a, b]$ is compact in $\mathbb{R}$.

(ii) Any $K \subset \mathbb{R}$ is compact iff it is bounded and closed.

For a proof, see Rudin (1976). From Proposition A.4.1, it is seen that the inverse image of an open set under a continuous function is open but the forward image may not have this property. But the following is true.

Theorem A.4.3: *Let $(\mathbb{S}, d)$ and $(\mathbb{T}, \rho)$ be two metric spaces and let $f : (\mathbb{S}, d) \to (\mathbb{T}, \rho)$ be continuous. Let $K \subset \mathbb{S}$ be compact. Then $f(K)$ is compact.*

The proof is left as an exercise (Problem A.35).

A.4.4 Sequences of functions and uniform convergence

Definition A.4.11: Let $(\mathbb{S}, d)$ and $(\mathbb{T}, \rho)$ be two metric spaces and let $\{f_n\}_{n\geq 1}$ be a sequence of functions from $(\mathbb{S}, d)$ to $(\mathbb{T}, \rho)$. The sequence $\{f_n\}_{n\geq 1}$ is said to:

(a) *converge pointwise to f on a set $A \subset \mathbb{S}$* if $\lim_{n\to\infty} f_n(x) = f(x)$ for each x in A;

(b) *converges uniformly to f on a set $A \subset \mathbb{S}$* if for each $\varepsilon > 0$, there exists $N_\varepsilon > 0$ (depending on ε and A) such that

$$n \geq N_\varepsilon \Rightarrow \rho\big(f_n(x), f(x)\big) < \varepsilon \quad \text{for all} \quad x \quad \text{in} \quad A.$$

A consequence of uniform convergence is the preservation of the continuity property.

Theorem A.4.4: *Let $(\mathbb{S}, d)$ and $(\mathbb{T}, \rho)$ be two metric spaces and let $\{f_n\}_{n\geq 1}$ be a sequence of functions from $(\mathbb{S}, d)$ to $(\mathbb{T}, \rho)$. Let for each $n \geq 1$, f_n be continuous on $A \subset \mathbb{S}$. Let $\{f_n\}_{n\geq 1}$ converge to f uniformly on A. Then f is continuous on A.*

Proof: The proof is based on the "break up into three parts" idea. By the triangle inequality,

$$\rho\big(f(x), f(y)\big) \leq \rho\big(f(x), f_n(x)\big) + \rho\big(f_n(x), f_n(y)\big) + \rho\big(f_n(y), f(y)\big).$$

Fix x in A. By the uniform convergence on A, $\sup\{\rho\big(f_n(u), f(u)\big) : u \in A\} \to 0$ as $n \to \infty$. So for each $\varepsilon > 0$, there exists $N_\varepsilon < \infty$ such that $n \geq N_\varepsilon \Rightarrow \rho\big(f_n(u), f(u)\big) < \frac{\varepsilon}{3}$ for all u in A. Now since $f_{N_\varepsilon}(\cdot)$ is continuous on A, there exists a $\delta > 0$ (depending on N_ε and x), such that $d(x, y) < \delta, y \in A \Rightarrow \rho\big(f_{N_\varepsilon}(y), f_{N_\varepsilon}(x)\big) < \frac{\varepsilon}{3}$. Thus, $y \in A$, $d(x, y) < \delta \Rightarrow \rho\big(f(x), f(y)\big) < \frac{2\varepsilon}{3} + \frac{\varepsilon}{3} = \varepsilon$. $\square$

A.5 Problems

A.1 Express the following sets in the form $\{x : x \text{ has property } p\}$.

(a) The set A of all integers which when divided by 7 leave a remainder ≤ 3.

(b) The set B of all functions form $[0, 1]$ to $\mathbb{R}$ with at most two discontinuity points.

(c) The set C of all students at a given university who are graduate students with at least one course in mathematics at the graduate level.

(d) The set D of all algebraic numbers. (A number x is called an *algebraic number*, if it is the root of a polynomial with rational coefficients.)

(e) The set E of all possible sequences whose elements are either 0 or 1.

A.2 Give an example of sets A_1, A_2 such that $A_1 \cap A_2 \neq A_1 \cup A_2$.

A.3 Let $I = [0, 1]$, $\Omega = \mathbb{R}$ and for $\alpha \in \mathbb{R}$, $A_\alpha = (\alpha - 1, \alpha + 1)$, the open interval $\{x : \alpha - 1 < x < \alpha + 1\}$.

(a) Show that $\cup_{\alpha \in I} A_\alpha = (-1, 2)$ and $\cap_{\alpha \in I} A_\alpha = (0, 1)$.

(b) Suppose $J = \{x : x \in I, x \text{ is rational}\}$. Find $\cup_{x \in J} A_x$ and $\cap_{x \in J} A_x$.

A.4 With $\Omega \equiv \mathbb{N} \equiv \{1, 2, 3, \ldots\}$, find A^c in the following cases:

(a) $A = \{\omega : \omega \text{ is divisible by 2 or 3 or both}\}$. If $\omega \in A^c$, what can be said about its prime factors?

(b) $A = \{\omega : \omega \text{ is divisible by 15 and 16}\}$.

(c) $A = \{\omega : \omega \text{ is a perfect square}\}$.

A.5 Show that $X \equiv \{0, 1\}^{\mathbb{N}}$, the set of all sequences $\{\omega_i\}_{i \in \mathbb{N}}$ where each $\omega_i \in \{0, 1\}$, is *uncountable*. Conclude that $\mathcal{P}(\mathbb{N})$ is uncountable.

A.6 Show that if Ω_i is countable for each $i \in \mathbb{N}$, then for each $k \in \mathbb{N}$, $\times_{i=1}^{k} \Omega_i$ is countable and $\cup_{i \in \mathbb{N}} \Omega_i$ is also countable but $\times_{i \in \mathbb{N}} \Omega_i$ is *not countable*.

A.7 Show that the set of all polynomials in x with integer coefficients is countable.

A.8 Show that the well ordering property implies the principle of induction.

A.9 Apply the principle of induction to establish the following:

(a) For each $n \in \mathbb{N}$, $\sum\limits_{j=1}^{n} j^2 = \frac{n(n+1)(2n+1)}{6}$.

(b) For each $n \in \mathbb{N}$, $x_1, x_2, \ldots, x_k \in \mathbb{R}$,

(i) (*The binomial formula*). $(x_1 + x_2)^n = \sum\limits_{r=0}^{n} \binom{n}{r} x_1^r x_2^{n-r}$.

(ii) (*The multinomial formula*).

$$(x_1 + x_2 + \ldots + x_k)^n = \sum \frac{n!}{r_1! r_2! \ldots r_k!} x_1^{r_1} x_2^{r_2} \ldots x_k^{r_k},$$

where the summation extends over all $(r_1, r_2, \ldots, r_k)$ such that $r_i \in \mathbb{N}$, $0 \leq r_i \leq n$, $\sum_{r=1}^{k} r_i = n$.

A.10 Verify that on $\mathbb{R}$, the relation $x \sim y$ if $x-y$ is rational is an equivalence relation but the relation $x \sim y$ if $x - y$ is irrational is not.

A.11 Show that the set $A = \{r : r \in \mathbb{Q}, r^2 < 2\}$ is bounded above in $\mathbb{Q}$ but has no l.u.b. in $\mathbb{Q}$.

A.12 Show that for any two sequences $\{x_n\}_{n\geq 1}$, $\{y_n\}_{n\geq 1} \subset \mathbb{R}$,

$$\varliminf_{n\to\infty} x_n + \varliminf_{n\to\infty} y_n \leq \varliminf_{n\to\infty} (x_n + y_n) \leq \varlimsup_{n\to\infty} (x_n + y_n)$$
$$\leq \varlimsup_{n\to\infty} x_n + \varlimsup_{n\to\infty} y_n.$$

A.13 Verify that $\lim\limits_{n\to\infty} x_n = a \in \mathbb{R}$ iff $\varliminf\limits_{n\to\infty} x_n = \varlimsup\limits_{n\to\infty} x_n = a$.

A.14 Establish Proposition A.2.3.

(**Hint:** First show that a Cauchy sequence is bounded and then show that $\varliminf\limits_{n\to\infty} x_n = \varlimsup\limits_{n\to\infty} x_n$.)

A.15 (a) Prove Proposition A.2.4 by comparison with the geometric series.

(b) Show that for integer $k \geq 1$, the power series $\sum_{n=k}^{\infty} n(n-1)(n-k+1)a_n x^{n-k}$ has the same radius of convergence as $\sum_{n=0}^{\infty} a_n x^n$.

A.16 Show that the series $\sum_{j=2}^{\infty} \frac{1}{j(\log j)^p}$ converges for $p > 1$ and diverges for $p \leq 1$.

A.17 Find the radius of convergence, ρ, for the powers series $A(x) \equiv \sum_{n=0}^{\infty} a_n x^n$ where

(a) $a_n = \frac{n}{(n+1)}$, $n \geq 0$.

(b) $a_n = n^p$, $n \geq 0$, $p \in \mathbb{R}$.

(c) $a_n = \frac{1}{n!}$, $n \geq 0$ (where $0! = 1$).

A.18 (a) Find the Taylor series at $a = 0$ for the function $f(x) = \frac{1}{1-x}$ in $I \equiv (-1, +1)$ and show that it converges to $f(x)$ on I.

(b) Find the Taylor series of $1 + x + x^2$ in $I = (1, 3)$, centered at 2.

(c) Let

$$f(x) = \begin{cases} e^{-\frac{1}{x^2}} & \text{if } \ |x| < 1, x \neq 0 \\ 0 & \text{if } \ x = 0 \ . \end{cases}$$

(i) Show that f is infinitely differentiable at 0 and compute $f^{(j)}(0)$ for all $j \geq 1$.

(ii) Show that the Taylor series at $a = 0$ converges but not to f on $(-1, 1)$.

A.19 Let $S = \{z : z \in \mathbb{C}, |z| = 1\}$ be the unit circle. Using the parameterization $t \to e^{\iota t} = (\cos t + \iota \sin t)$ from $[0, 2\pi]$ to S, show that the arc length of S (i.e., the circumference of the limit circle) is 2π.

A.20 Set $\phi(t) = \frac{\sin t}{\cos t}$ for $-\frac{\pi}{2} < t < \frac{\pi}{2}$. Verify that $\phi' = 1 + \phi^2$ and that $\phi : (-\frac{\pi}{2}, \frac{\pi}{2})$ to $(-\infty, \infty)$ is strictly monotone increasing and onto. Conclude that

$$\int_{-\infty}^{\infty} \frac{1}{1+x^2} dx = \int_{-\pi/2}^{\pi/2} \frac{\phi'(t)}{1 + (\phi(t))^2} dt = \pi.$$

A.21 Using the property that $e^{\iota \frac{\pi}{2}} = \iota$ verify that for all t in $\mathbb{R}$

$$\begin{aligned} &\cos(\frac{\pi}{2} - t) = \sin t, \ \sin(\frac{\pi}{2} - t) = \cos t, \\ &\cos(\pi + t) = -\cos t, \ \sin(\pi + t) = -\sin t, \\ &\cos(2\pi + t) = \cos t, \ \sin(2\pi + t) = \sin t. \end{aligned}$$

Also show that $\cos t$ is a strictly decreasing map from $[0, \pi]$ onto $[-1, 1]$ and that $\sin t$ is a strictly increasing map from $[-\frac{\pi}{2}, \frac{\pi}{2}]$ onto $[-1, 1]$.

A.22 Using (i) of Theorem A.3.1, express $\cos(t_1 + t_2)$, $\sin(t_1 + t_2)$ in terms $\cos t_i$, $\sin t_i$, $i = 1, 2$ and in turn use this to prove Corollary A.3.4 from Theorem A.3.3.

A.23 Verify that $p_n(z) \equiv (1 + \frac{z}{n})^n$ converges to e^z uniformly on bounded sets in $\mathbb{C}$.

A.24 (a) Verify that for $p = 1$, $p = 2$ and $p = \infty$, d_p is a metric on $\mathbb{R}^k$.

(b) Show that for fixed x and y, $\varphi(p) \equiv d_p(x, y)$ is continuous in p on $[1, \infty]$.

(c) Draw the open unit ball $B_p \equiv \{x : x \in \mathbb{R}^2,\ d_p(x,0) < 1\}$ in $\mathbb{R}^2$ for $p = 1, 2$ and ∞.

A.25 Let $\mathbb{S} = C[0,1]$ be the set of all real valued continuous functions on $[0,1]$. Now let

$$\begin{aligned} d_1(f,g) &= \int_0^1 |f(x) - g(x)|dx, \quad \text{(area metric)} \\ d_2(f,g) &= \left(\int_0^1 |f(x) - g(x)|^2 dx\right)^{\frac{1}{2}}, \quad \text{(least square metric)} \\ d_\infty(f,g) &= \sup\{|f(x) - g(x)| : 0 \le x \le 1\} \quad \text{(sup metric).} \end{aligned}$$

Show that all these are metrics on $\mathbb{S}$.

A.26 Let $\mathbb{S} = \mathbb{R}^\infty \equiv \{\{x_n\}_{n\ge1} : x_n \in \mathbb{R}$ for all $n \ge 1\}$ be the space of all sequences of real numbers. Let $d(\{x_n\}_{n\ge1}, \{y_n\}_{n\ge1}) = \sum_{j=1}^\infty \left(\frac{|x_j - y_j|}{1+|x_j-y_j|}\right)\frac{1}{2^j}$. Show that $(\mathbb{S}, d)$ is a Polish space.

A.27 If $s_k = \{x_{kn}\}_{n\ge1}$ and $s = \{x_n\}_{n\ge1}$, are elements of $\mathbb{S} = \mathbb{R}^\infty$ as in Problem A.26, verify that as $k \to \infty$, $s_k \to s$ iff $x_{kn} \to x_n$ for all $n \ge 1$.

A.28 Establish Theorem A.4.1.

A.29 Let $\mathbb{S} = C[0,1]$ and $d_p(f,g) \equiv \left(\int_0^1 |f(t) - g(t)|^p dt\right)^{\frac{1}{p}}$ for $1 \le p < \infty$ and $d_\infty(f,g) = \sup\{|f(t) - g(t)| : t \in [0,1]\}$.

(a) Let $f(x) \equiv 1$. Let $f_n(t) \equiv 1$ for $0 \le t \le 1 - \frac{1}{n}$, and $f_n(t) = n(1-t)$ for $1 - \frac{1}{n} \le t \le 1$. Show that $d_p(f_n, f) \to 0$ for $1 \le p < \infty$ but $d_\infty(f_n, f) \not\to 0$.

(b) Fix $f \in C[0,1]$. Let $g_n(t) = f(t)$, $0 \le t \le 1 - \frac{1}{n}$, and $g_n(t) = f(1 - \frac{1}{n}) + (f(1) - f(1 - \frac{1}{n}))n(t + \frac{1}{n} - 1)$, $1 - \frac{1}{n} \le t \le 1$.

Show that $d_p(g_n, f) \to 0$ for all $1 \le p \le \infty$.

A.30 Show that if $\{x_n\}_{n\ge1}$ is a convergent sequence in a metric space $(\mathbb{S}, d)$, then it is *Cauchy*.

A.31 Verify (b) of Example A.4.2 from the axioms of real numbers (cf. Proposition A.2.3). Verify (c) of the same example from (b).

A.32 Let $\mathbb{S} = C[0,1]$ and d be the *supremum metric*, i.e.,

$$d(f,g) = \sup\{|f(x) - g(x)| : 0 \le x \le 1\}.$$

By approximating any continuous function with piecewise linear functions with rational end points and rational values, show that $(\mathbb{S}, d)$ is Polish, i.e., it is complete and separable.

A.33 Show that the function $f(x) = x^2$ is continuous on $\mathbb{R}$, uniformly so on any bounded set $B \subset \mathbb{R}$ but not uniformly on $\mathbb{R}$.

A.34 Show that unions of open sets are open and intersection of any two open sets is open. Give an example to show that the intersection of an infinite number of open sets need not be open.

A.35 Prove Theorem A.4.3.

A.36 Let $f_n(x) = x^n$ and $g(x) \equiv 0$ on $\mathbb{R}$. Then $\{f_n\}_{n\geq 1}$ converges pointwise to g on $(-1, 1)$, uniformly on $[a, b]$ for $-1 < a < b < 1$, but not uniformly on $(0, 1)$.

A.37 Let $\{f_n\}_{n\geq 1}$, $f \in C[0, 1]$. Let $\{f_n\}_{n\geq 1}$ converge to f uniformly on $[0, 1]$. Show that $\lim_{n\to\infty} \int_0^1 |f_n(x) - f(x)|dx = 0$ and $\lim_{n\to\infty} \int_0^1 f_n(x) = \int_0^1 f(x)dx$.

A.38 Give a proof of Proposition A.2.6 (vi) (term by term differentiability of a power series) using Proposition A.2.7 (iv) (the fundamental theorem of Riemann integration).